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Nannan Li

Faculty member | INQUA Fellow

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Education

PhD (Doctorate) in Physical Geography Northeast Normal University (China) 2015.09-2020.06
PhD dissertation: *Responses of vegetation to past climate changes since the Younger Dryas: investigation from Gushantun peatland in Northeastern China* (**Supervisor:** Prof. Dongmei Jie)
Visiting PhD student CEREGE (France) *Triple oxygen isotope analysis of Phytoliths* 2018.10-2019.09
Visiting PhD student Eawag (Switzerland) *Biomarkers in Paleolimnology* 2018.04-2018.06
Bachelor in Geography (GPA: 3.72/4) Northeast Normal University (China) 2010.09-2014.06

Work Experience

Lecture Department of Geography, Fujian Normal University 2026.9-Present
Postdoctoral Researcher Department of Geography, Maynooth University 2023.12-2026.9
INQUA Fellow Swiss Federal Institute of Aquatic Science and Technology(Eawag) 2023.03-2023.07
Postdoctoral Fellow Xiamen University 2020.09-2023.11

Research interests and Skills

Interests Late Quaternary paleoclimatology and paleoecology, Geochemistry, and global change
Analytical Skills Phytolith analysis, (silicates) triple oxygen isotope analysis, radiocarbon dating, stable carbon and oxygen isotopes, diatom extraction, grain size, humification, bulk geochemistry analysis, time series analysis
Field Sampling Topsoil sampling, peat corer (Eijkelkamp), piston corer
Lab Equipment Microtrac S3500 laser particle size analyzer, Olympus biologic microscope (BX53), OI Total Organic Carbon (TOC) analyzer 1030S, Radiocarbon dating graphite preparation, EA-IRMS, Avaatech XRF Core Scanner
Programming Language R
Language English (Fluent), French (Beginner), Chinese (Native)

Funding and Awards

Funding
2024 Irish Quaternary Association (IQUA) Research Awards **EUR 1,200**
2024 Geological Survey Ireland (GSI) Short Call 2024 **EUR 29,979**
2024 Research Incentivisation Award 2024 (Department of Geography, Maynooth University) **EUR 3,880**
2022 INQUA Fellowship 2023 **EUR 15,000**
Biomarker insights towards an improved understanding of Northeastern Asian monsoon during the Younger Dryas
2022 Open Research Foundation in the Key Laboratory of Geographical Processes and Ecological Security in Changbai Mountains, Ministry of Education of China **CNY 50,000**
Quantitative reconstruction of Holocene relative humidity (RH) in the Changbai Mountains, NE China
2021 Chinese Postdoctoral Science Foundation **CNY 80,000**
Impacts of human disturbance and climate change on late-Holocene vegetation dynamics in SE China
2020 Outstanding Postdoctoral Fellowship at MEL, Xiamen University **CNY 720,000**
2018 International Exchange program for Excellent PhD students in China **EUR 16,200**
2017 International Exchange program for Excellent Youth in China **CNY 50,000**

Awards
2020 Presidential Scholarship of Northeast Normal University (awarded to the top 5% of students) **CNY 10,000**
2018 Chinese National Scholarship for PhD students (awarded to the top 1% Chinese PhD students) **CNY 30,000**
2018 NENU-Person of the Year 2018 (awarded to ten excellent students per year at Northeast Normal University) **CNY 8,000**
2013 Presidential Scholarship of Northeast Normal University (awarded to the top 5% of students) **CNY 3,000**
2013 Baogang scholarship of China (awarded to the top 0.1% of students in China) **CNY 8,000**
2012 Presidential Scholarship of Northeast Normal University (awarded to the top 5% of students) **CNY 3,000**
2011 Chinese National Scholarship for Bachelor students (awarded to the top 1.5% Bachelor students) **CNY 8,000**

Honors
2018 NENU-Metal (awarded to ten excellent Bachelor students per year at Northeast Normal University)
2018 Excellent College Students of Changchun City (awarded to the top 1% students per year in the Changchun city)

Peer-reviewed Publications

Papers published in international journals (selected)

2023 Yu, F.*, **Li, N.***, Tian, G., Huang, Z., Xiong, H., Li, T., Liu, S. and Liu, Y., 2023. A re-evaluation of Holocene relative sea-level change along the Fujian coast, southeastern China. ***Palaeogeography, Palaeoclimatology, Palaeoecology*** 622: 11577.

2023 **Li, N.**, Yu, F., Chambers, F., Huang, Z., Lin, W., Zhu, Z., Yang, H., and Lin, J., 2023. A humification-based method towards refining Holocene radiocarbon chronologies: wetland records from southeastern China. ***The Holocene*** 33: 605-615.

2022 **Li, N.**, Yu, F., Sack, D., Huang, Z., Tian, G., and Liu, S., 2022. Phytolith assemblages as a promising tool for quantitative canopy coverage reconstruction in subtropical forests, China. ***Frontiers in Plant Science*** 13: 912627.

Li, N., Song, L., Sack, D., Lu, Z., Yu, F., Gao, G., Li, D., Li, M., Yang, Y., Zong, Y., and Jie, D., 2022. Phytolith and simulation evidence for precipitation-modulated vegetation dynamics along the East Asian monsoon margin. ***Palaeogeography, Palaeoclimatology, Palaeoecology*** 590: 110842.

Gao, G., Rand, E., **Li, N.**, Li, D, Wang, J., Niu, H., Meng, M., Liu, Y., and Jie, D., 2022. East Asian monsoon modulated Holocene spatial and temporal migration of forest-grassland ecotone in Northeast China. ***Catena*** 213: 106151.

2021 **Li, N.**, Sharifi, A., Chambers, F.M., Ge, Y., Dubois, N., Gao, G., Li, D., Liu, L., Liu, H., Wang, J., Niu, H., Meng, M., Liu, Y., Zhang, G., and Jie, D. Linking Holocene East Asian monsoon variability to solar forcing and ENSO activity: Multi-proxy evidence from a peatland in Northeastern China. ***The Holocene*** 31: 966-982.

Li, N., Xie, M., Sack, D., Dubois, N., Yang, X., Gao, G., Li, D., Liu, L., Liu, H. and Jie, D, 2021. Continuous aridification since the mid-Holocene as the main cause of C3/C4 dynamics in the grasslands of northeastern China. ***European Journal of Soil Science*** 72: 356-371.

Gao, G., Jie, D., Li, D., **Li, N.**, Liu, L., Liu, H., Wang, J., Niu, H., Meng, M., Liu, Y., and Zhang, G., 2021. Reconstruction of local-scale plant community characteristics using phytolith: A case study of a forested region in Northeast China. ***Catena*** 203: 105146.

2020 **Li, N.**, Li, M., Sack, D., Kang, W., Song, L., Yang, Y., Zong, Y. and Jie, D, 2020. Diatom evidence for mid-Holocene peatland water-table variations and their possible link to solar forcing. ***Science of the Total Environment*** 725: 138272.

Li, N., Sack, D., Sun, J., Liu, S., Liu, B., Wang, J., Gao, G., Li, D., Song, Z. and Jie, D., 2020. Quantifying the carbon content of aeolian sediments: Which method should we use? ***Catena*** 185: 104276.

2019 Gao, G., Jie, D., Li, D., **Li, N.**, Liu, L., Liu, H., Leng, C., Wang, J., Liu, B. and Li, P., 2019. Assessing modern arboreal phytolith sensitivity to vegetation variations in temperate forest regions. ***BOREAS*** 48: 731–745.

2018 Gao, G., Jie, D., Li, D., **Li, N.**, Liu, L., Liu, H., Shi, J., Leng, C. and Wang, J., 2018. Reliability of phytoliths for reconstructing vegetation dynamics in northern temperate forest regions: A case study in northeast China. ***Quaternary Science Reviews*** 201: 1–12.

Li, N., Sack, D., Gao, G., Liu, L., Li, D., Yang, X., Jie, D., Liu, H., Shi, J. and Leng, C., 2018. Holocene *Artemisia*-Chenopodiaceae-dominated grassland in North China: Real or imaginary? ***The Holocene*** 28: 834–841.

2017 **Li, N.**, Chambers, F.M., Yang, J., Jie, D., Liu, L., Liu, H., Gao, G., Gao, Z., Li, D., Shi, J., Feng, Y., and Qiao, Z., 2017. Records of East Asian monsoon activities in Northeastern China since 15.6 ka, based on grain size analysis of peaty sediments in the Changbai Mountains. ***Quaternary International*** 447: 158–169.

2014 **Li, N.**, Jie, D., Ge, Y., Guo, J., Liu, H., Liu, L. and Qiao, Z., 2014. Response of phytoliths in *Phragmites communis* to elevated CO_2 concentration in Songnen Grassland, China. ***Quaternary International*** 321: 97–104.

Presentations at professional meetings

2024 **Oral Presentation at the 35th International Geographical Congress (IGC)** **Dublin**, August 25-30, 2024

Li, N., Shaw, H., Ryan, C., Orme, L. The Influence of Past Climate Change on the Carbon Accumulation Rates of Blanket Bogs (PCARB): An Irish Perspective.

2023 **Oral Presentation at the XXI INQUA Congress** **Rome**, July 14-20, 2022

Li, N., Yu, F. Can Poaceae pollen serve as a reliable proxy of human disturbance in subtropical forests?—Insights from palynology and geochemistry perspectives.

2022 **Oral Presentation at the Pre IAL-IPA 2022 Early Career Virtual Meeting** **Online**, April 26-29, 2022

Li, N., Yu, F. Can Poaceae pollen serve as a reliable proxy of human disturbance in subtropical forests?—Insights from palynology and geochemistry perspectives.

2021 **Poster Presentation at 6th Conference on Earth System Science** **Shanghai**, October 18-21, 2021

Li, N., Song, L., Lu, Z., Yu, F., Gao, G., and Jie, D. Phytolith and simulation evidence for precipitation-modulated vegetation dynamics along the East Asian monsoon margin.

2021 **Oral Presentation at the 7th International Conference on Estuaries and Coasts** **Shanghai**, July 6-10, 2021

Li, N., Du, F., Yu, F., and Jie, D. Sea-level and monsoon climate co-modulated sedimentary environment evolution in the Liaodong Bay, China—A phytolith perspective.

2019 **Poster Presentation at 26th congress of PhD students in Environmental sciences** **Marseille**, April 29-30, 2019

Li, N., Alexandre, A., Couapel, M., Sonzogni, C., Mazur, J.-C., Outrequin, C. and Jie, D. Phytolith ^{17}O-excess as a proxy for past relative humidity reconstruction: calibration using *Phragmites australis* (common reed) along a relative humidity gradient in Northeastern China.

2018 **Oral Presentation at the 11th International Meetings on Phytolith Research** **Wuhan**, September 14-16, 2018

Li, N., Jie, D., Xie, M., Gao, G., Liu, L., Liu, H., Li, D. and Yang, X. Holocene C3/C4 Abundance Dynamics in Songnen Grasslands and Its Link to Monsoon-driven Climate Variations.

2017 **Oral Presentation at the 6th National Conference of Sedimentology in China** **Nanjing**, October 27-30, 2017
Li, N., Jie, D., Gao, G., Liu, L., Li, D. and Yang, X. Holocene paleovegetation reconstructions in Songnen Sandy Lands: *Artemisia*-Chenopodiaceae dominated or Poaceae dominated?

Teaching and Mentoring Experiences

Guest Lecture

2024.12 **Introduction to Palaeoecology**——Maynooth University
(Lecture: *Application of R for Quantitative Analyses in Palaeoecology)*

2022.11 **Reconstructing Earth's Climate History**——Xiamen University
(Lecture: *Introduction of Phytolith Analysis)*

2016.11 **An Introduction to Geology: Practical Classes**——Northeast Normal University
(Lectures: *Classification, Identification, and Description of Igneous Rocks and Metamorphic Rocks)*

2016.05 **Reconstructing Quaternary Environments: Practical Classes**——Northeast Normal University
(Lectures: *An Introduction to Grain Size Analysis and Its Applications in Quaternary Research)*

Mentoring

2021-2023 **National Innovation Training Program for Undergraduate Students**
Yang, S. *The reconstruction of Holocene Sea-level change and coastal lines in the Bohai Sea*

2021-2022 **Innovation Training Program for Bachelor Students (Xiamen University)**
Zeng, Y., Tan, X. and Huang Y. *The potential of using phytolith assemblages for quantitative canopy coverage reconstruction in subtropics: modern calibrations in subtropical forests in China*

2021-2022 **Innovation Training Program for Bachelor Students (Xiamen University)**
Zhu, X., Shu, Y. and You, Y. *Phytoliths in modern plants from the National Natural Reserve of the Wuyi Mountains and their implications for paleovegetation reconstruction*

2020-2021 **Innovation Training Program for Bachelor Students (Xiamen University)**
Lin, W., Zhu, Z., Yang, H. and Lin, J. *Paleoclimate and paleovegetation reconstructions for the subtropics in China: A multiproxy-based wetland record from the Daiyun Mountains*

Analytical Experiences

- Phytolith assemblage analysis (including chemical extraction, microscope identification, and counting)
- Phytolith (silicates) triple oxygen isotope analysis using IR laser-heating fluorination technique
- Stable isotope analyses of C and N in bulk organic materials (EA-IRMS)
- Radiocarbon dating, including the selection of macrofossils, chemical preparations for various materials (bulk organic matters, macrofossils, charcoal, and shells), and graphite preparation using the vacuum line
- Grain size analysis and mathematical methods used for different grain size components' decomposition
- Geochemical analysis, including humification degree, TOC, carbonates, free iron, and heavy metals concentrations using ICP-MS or visible spectrophotometer
- Biomarker (particularly leaf wax) analysis, chemical extraction, and purification of different compounds
- Peat and lake sediments coring, splitting sediment cores, sedimentary description, and sub-sampling
- Time-series analysis including wavelet, spectrum, and Empirical Mode Decomposition (EMD)

Field Experiences

- Peat sediment coring in 30 blanket bogs across the Republic of Ireland
- Sediment coring on the coastal plain of the Guangdong province in southern China
- Peat coring in five different peatlands in NE and SE China
- Modern plant and topsoil sampling across a variety of natural environmental gradients across China
- Surface sediments sampling along the sea-level transect in the Mangrove forests, SE China
- Six long field survey and sampling journeys (>500 km) across northeastern China and on five of those, I was the primary organizer and responsible for the scientific communication and logistics
- Lake sediment coring using gravity and Piston corer in Swiss lakes (Lake Zürich, Lake Murten)
- Water sampling prepared for H, O, and C isotope analysis (δD, δ^{18}O, δ^{13}C) across an extensive area in northeastern China

Peer reviewing and Professional memberships

- **Guest editor** of a special volume in *Review of Palaeobotany and Palynology*, (SI: Phytoliths in Asia) 2021.11-present
- **Peer reviewer** for *the Holocene, Annals of Botany, Journal of Hydrology, Sciences in Cold and Arid Regions, Review of Palaeobotany and Palynology,Frontiers in Earth Science, Journal of Archaeological Science: Reports* 2017.10-present
- **Permanent Member** of the Chinese Association for Quaternary Research (CHIQUA) 2017.05-present
- Member of the Irish Quaternary Association (IQUA) 2024.01-2026.12
- Member of the British Ecological Society (BES) 2023.08-2024.08